



Habitat use and tolerance levels of macroinvertebrates concerning hydraulic stress in hydropeaking rivers – A case study at the Ziller River in Austria

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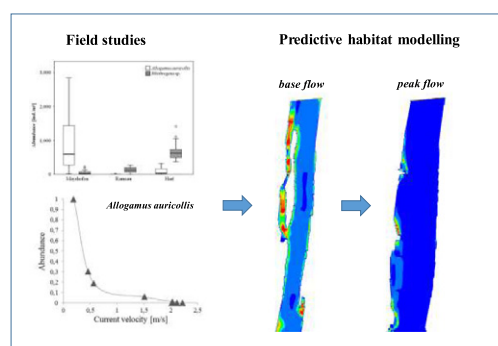
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HIGHLIGHTS

- Habitats of stagnophilic macroinvertebrate taxa are significantly minimized in channelized stretches affected by hydropeaking.
- The WFD compliant national Austrian assessment method fails to detect impacts of hydropeaking on macroinvertebrates.
- The development of a stressor-specific sampling design is required as the MHS largely ignores vulnerable habitats.
- The hydraulic stress analysis provides expertise on the resistance of certain taxa in terms of hydropeaking.

GRAPHICAL ABSTRACT



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ABSTRACT

Artificial flow fluctuations due to the operation of hydropower plants, frequently described as hydropeaking, result in a constant decrease of biomass of specific macrozoobenthos (MZB) taxa. For the presented case study, we assessed three reaches in the Ziller River catchment. At each sampling reach we performed the Multi-Habitat-Sampling (MHS) method with a Water Framework Directive (WFD) compliant AQEM/MHS net according to the Austrian guideline. Additionally, a hydraulic-specific sampling was conducted with a modified Box (Surber) sampler. As a basis for predictive habitat modelling of the MZB fauna, we measured abiotic parameters like mean (v_{40}) and bottom-near (v_{bottom}) flow rate or water depth respectively, for each box sample. In addition, the choriotope type, representing grain size classes, was determined. One of the main results is, that the national status assessment was not capable to reflect the impact of pulse release at the investigated river stretches on the basis of status classes. Moreover, we figured out that 1) habitats of stagnophilic macroinvertebrate taxa are minimized in channelized stretches affected by hydropeaking, leading to heavy quantitative losses for populations, becoming apparent in significant decreases in total individual numbers and biomass for many taxa. 2) The minor respond of the ecological status class in affected stretches by applying the WFD compliant national assessment method for macroinvertebrates owes to the tolerance of rheobiont or rheophilic taxa commonly classified

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as indicators for good conditions regarding saprobity or degradation score. 3) A development of a stressor-specific sampling design is required as the MHS method largely ignores vulnerable habitats. 4) The habitat suitability of selected species provides efficient expertise for impact assessment and mitigation measure design in terms of predictive habitat modelling.

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1. Introduction

Artificial flow fluctuations due to the operation of hydropower plants, frequently described as hydropeaking (Moog, 1993; Saltveit et al., 2001), lead to an increase of physiological stress (Valentin et al., 1995), accelerating macroinvertebrate's (MZB) drift rate (e.g. Meile et al., 2005; Limnex, 2009) resulting in a constant decrease of biomass of specific taxa (Schweizer et al., 2009). In addition, changes in substrate composition and sediment quality are associated with hydropeaking (Anselmetti et al., 2007; Hauer et al., 2014) and reservoir operation in general (Poff and Hart, 2002; Wohl and Cenderelli, 2010). According to Wood and Armitage (1997) fine sediment deposition additionally impairs the MZB fauna by changing the substrate composition (Richards and Bacon, 1994), generally resulting in a reduction of abundance and diversity (e.g. Culp et al., 1985). Hence, both hydraulic stress and changes in substrate composition are crucial drivers of macroinvertebrate population in rivers with hydropeaking impacts.

Concerning the aims of the European Water Framework Directive (to achieve and preserve the good ecological status) the two aspects (i) sampling design and (ii) management tools turned out to be important and adapted for the analysis and the management of artificial flow fluctuation by hydropower on macroinvertebrates. The Water Framework Directive (WFD) compliant method for assessing the ecological status of rivers, concerning the Biological Quality Element macroinvertebrates, is exclusively based on the principle of the Multi-Habitat-Sampling (MHS) providing two assessment modules, Organic Pollution (saprobity) and General Degradation (multi-metric index) in Austria (Ofenböck et al., 2010). Previous studies (e.g. Graf et al., 2013) dealing with the effects of hydropeaking on benthic invertebrates already indicated that existing methods do not always lead to convincing results as abundance of populations is excluded from the analyses. Alpine rivers in Central Europe, with a few exceptions, are uniformly straightened and channelized, showing a general increase of flow velocity and hydraulic stress. Considering the hypothesis that lentic species inhabiting bank structures or habitats along the shoreline are worst affected by the additive stressor hydropeaking, we designed and applied an individual sampling design for this qualitative and quantitative survey.

For the required management and the need to implement predictive tools into the analysis of impacted rivers, habitat modelling turned out as a successful tool for a quantitative determination of (artificial) flow fluctuations (e.g. Tharme, 2003). Univariate functions, representing the habitat use related to a specific environmental variable (e.g. flow-velocity) were frequently applied in e-flow assessments based on habitat modelling (Ahmadi-Nedushan et al., 2006; Paredes-Arquiola et al., 2011). Univariate (suitability) functions for macroinvertebrates have been derived on hemisphere measurements (Bockelmann et al., 2004; Lancaster and Hildrew, 1993) representing near-bottom hydraulics (e.g. bottom shear stress), flow-velocity measurements (Shearer et al., 2015; Kelly et al., 2015), and Froude-number (Brooks et al., 2005; Jowett, 1993). However, studies about the habitat use of macroinvertebrates in terms of hydropeaking are missing. There is a lack of knowledge, which physical conditions are suitable under the restriction of artificial fluctuations in flow and consequently in altering hydraulics (e.g. flow velocities, bottom shear stress).

Thus, the aim of the present short communication is (i) to evaluate the national standard of macroinvertebrate sampling in terms of hydropeaking and (ii) to derive functional relationships between the abiotic environment (under the restriction of hydropeaking) and

habitat use (suitability) of selected macroinvertebrate species. The derivation of suitability curves for *Rhithrogena* sp. and *Allogamus auricollis* is referred to on-site flow velocity measurements of depth-averaged flow velocities (ms^{-1}) and water depth (m). The hydropeaking impact was investigated on the basis of a calibrated depth-averaged hydrodynamic numerical model.

2. Study reach

For the present case study, we selected three reaches in the Ziller River catchment. The stream is a right tributary of the river Inn with a length of 55.7 km and a total catchment of 1135 km². The spring area is located at an altitude of 2270 m·a.s.l., the altitude at the confluence with the river Inn in Strass is at 517 m·a.s.l. According to Illies (1978) the river belongs to the ecoregion “Alps”, with its source in the bioregion “glaciated Crystalline Alps”, the investigated reaches are, however, located in the bioregion “non-glaciated Crystalline Alps” (Moog et al., 2001). Hence, the Ziller River has a nivo-glacial regime with a runoff maximum in July (Mader et al., 1996). According to Wimmer (1994), the stream order at the confluence is 6.

The first reach in the river continuum (Mayrhofen) was surveyed as a non-impacted reference site (global coordinates: 47.32886 N, 11.87256 E) and the stretches Ramsau (global coordinates: 47.20524 N, 11.86927 E) and Hart (global coordinates: 47.32928 N, 11.86657 E) were examined for hydropeaking impacts on macroinvertebrates due to the operations of hydropower-plant (HPP) Mayrhofen and HHP Gerlos (Fig. 1). The Ziller River exhibits significant morphological changes resulting from flood protection measures during the 1960's. Hence, only straight and/or plane-bed morphology was present, with average bed slopes of 0.0043 and 0.0054 for the hydropeaking stretches (Table 1). However, we recorded local sedimentological deposits (initial bar formations) within groin fields (Hauer et al., 2014).

3. Methods

At each sampling reach ($n = 3$) the MHS method was performed with a WFD compliant AQEM/MHS net (AQEM Consortium, 2002) according to the Austrian guideline (Ofenböck et al., 2010). The MHS method focuses on a multi-habitat scheme designed for sampling major habitats proportionally according to their presence (of at least 5% coverage) within a sampling reach, providing 20 sub-samples pooled to a total sample. Additionally, a hydraulics-specific sampling was performed with a modified Box (Surber) sampler. In total we took 18 box samples per investigation-site separately in different distances from the shoreline to cover the hydraulic gradient. Both methods were performed during base-flow to characterize and compare the benthic invertebrate fauna of an unaffected (reference) stretch and two downstream stretches of the Ziller River impaired by consistently pulse release. Per sample all benthic invertebrates were sorted and counted, determined to the highest possible level (for MHS) or to the mixed ‘screening level’ (for box samples) according to Ofenböck et al. (2010). Biomass was ascertained only for box samples.

As a basis for the modelling of the hydraulic impact on the MZB fauna, abiotic parameters like mean (v_{40}) and bottom-near (v_{bottom}) flow rate (using FloMate Marsh McBirney 2000) and water depth respectively, were measured for each box sample. In addition, we determined the choriotope type (according to Moog et al., 1999), representing different grain size classes (Bunte and Abt, 2001). Due to

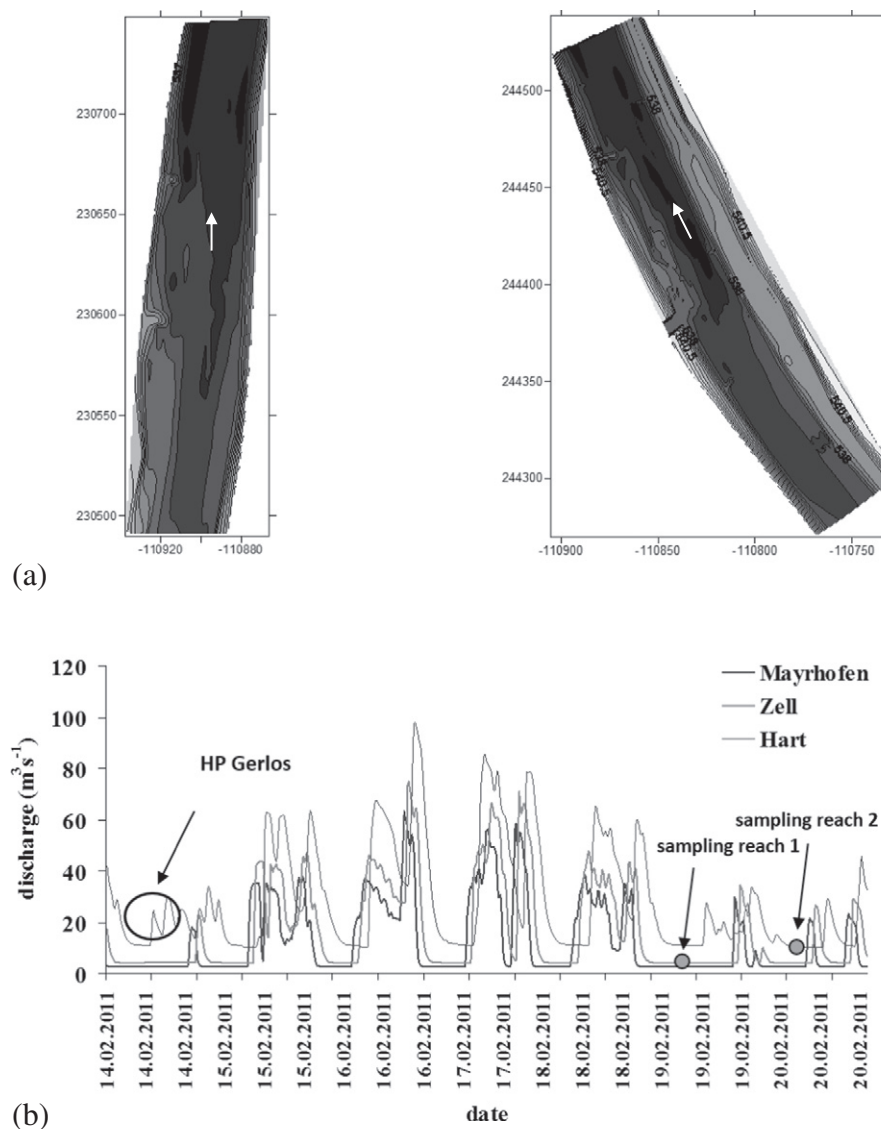


Fig. 1. (a) Digital Terrain Models (DTMs) of the investigated reaches at the Ziller River; Ramsau (left) and Hart (right); flow direction is indicated by white arrows; (b) Hydrographs of various gauging stations ($n = 3$) along the investigated Ziller River which exhibits the impact on flow by HP Mayrhofen and/or the possible superposition due to HP Gerlos (gauging station Hart); additionally, sampling dates are included in the Figure; HP = hydropower plant.

the varying abundance of taxa in specific hydraulic conditions (e.g. flow velocities, substrate), we created habitat suitability curves referring to the frequency of a taxon colonizing various habitat types: the ratio of habitat use is transformed into a measure of habitat selection (Manly et al., 2002). Habitat suitability values are standardized, ranging from 0 to 1 (Raleigh et al., 1986), whereas the highest number of individuals in a certain hydraulic class (e.g. velocity class) is estimated to indicate the highest suitability and is standardized to value 1; if this taxon is missing at a certain hydraulic class, value 0 is indicated. This

acquisition can be directly used in so called habitat selection models (also referred to as 'preference or habitat index' models or simply 'habitat models') which are widely used to evaluate habitat quality and the prediction of habitat alterations on faunal populations (Railsback et al., 2003).

The habitat suitability of both target taxa, *Allogamus auricollis* and *Rhithrogena* sp., enables a predictive habitat modelling based on depth-averaged two-dimensional unsteady flow hydraulics. Two sites of the Ziller River have been selected for hydrodynamic-numerical

Table 1
Description of the selected modelling reaches based on morphological, hydrological and sedimentological characteristics; Q_L = mean annual low-flow, Q_M = mean flow, D_{bf} = bankfull depth, W_{bf} = bankfull width, slope = average bed slope of the selected reach, length = total reach length used for modelling, bed_min = minimum bed elevation within the study reach, morph. = morphological characteristics, peak_flow = operating peaking mode of one or more turbines (Hauer et al., 2014).

No.	Site	Q_L (m³ s⁻¹)	Q_M (m³ s⁻¹)	D_{bf} (m)	W_{bf} (m)	Slope (—)	Length (m)	Bed_min (m a.s.l.)	Morph. (—)	Peak_flow (m³ s⁻¹)
1	Ziller_1 (n = 19)	8.90	43.30	5.2	57.4	0.0043	281.9	535.65	plane ¹⁾	6 × 15 + 28
2	Ziller_2 (n = 15)	3.90	29.30	3.4	49.3	0.0054	257.2	584.56	plane ¹⁾	6 × 15

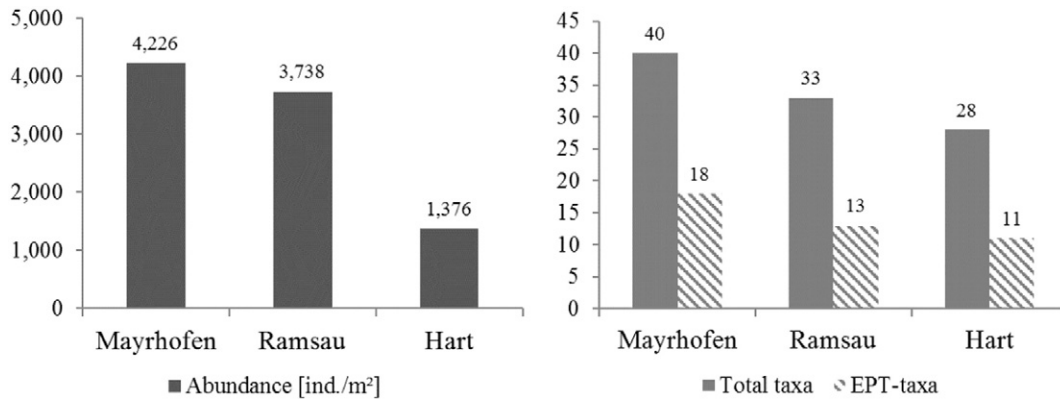


Fig. 2. Total abundance (left), number of total taxa and EPT (Ephemeroptera, Plecoptera, Trichoptera)-taxa (right) per sampling site; MHS samples.

(HN)-modelling: (i) for the first impacted study site in Ramsau (reach 1) a non-linear quadratic mesh (13,023 elements/12,995 nodes; spatial resolution: ~ 1 m) was calculated on the basis of linear interpolation of channel bathymetry (23 cross sections sampled by terrestrial survey; Leica TC805) (Fig. 1a). The same level of accuracy was achieved for the second impacted river stretch Hart (reach 2) based on a triangulated mesh of (35,759 elements/18,288 nodes; spatial resolution: ~ 1 m) (Fig. 1a). For both modelling stretches the SMS-software 9.0 (Boss International Surface Water Modelling System) was used as pre- (e.g. derive of modelling mesh) and post-processing tool (quantification and visualisation of depth-averaged flow distribution). The hydrodynamic – numerical modelling was performed with the Hydro-as-2d software (Nujic, 1999). Roughness coefficients were calibrated for base-flow; all details regarding this calibration and model performance are presented in Hauer et al. (2014) and are not mentioned here. Thus, based on time resolution of 15 min (hydrological input data) the total number of time steps ($n = 671$) was calculated for both modelling reaches (Fig. 1b). Base-flow was determined and analysed related to the measured low flow winter discharge during macroinvertebrate sampling in February 2011 ($3.95 \text{ m}^3 \text{ s}^{-1}$).

4. Results

The maximum peak-flow (gauging station Ramsau) of the last event prior to sampling reached $74.72 \text{ m}^3 \text{ s}^{-1}$ (Fig. 1c), which is >18 times higher than base-flow.

According to the WFD compliant Austrian assessment method, the reference stretch Mayrhofen, and the first affected stretch in the river continuum (Ramsau) show similar saprobity classes, indicating high status. The benthic community at the second impacted stretch near Hart indicates good status. The module General Degradation specifies

moderate status for all three stretches (the reference stretch only narrowly missed good status), thus, following the ‘worst case’ principle of the assessment method, all stretches indicate a moderate ecological status.

In Fig. 2, a distinct decrease in abundance and diversity regarding MHS samples is visible. At the second affected reach (Hart) the impact is most evident, resulting in an abundance loss of about two thirds (declining from 4226 to $1376 \text{ ind} \cdot \text{m}^{-2}$). Moreover, there is a diversity loss of one third compared to reference conditions, declining from 40 to 28 total taxa and from 18 to 11 EPT (Ephemeroptera, Plecoptera and Trichoptera) -taxa respectively. With regard to Box samples, abundance and biomass show significant ($p < 0.01$) declines from averaged $6700 \text{ ind} \cdot \text{m}^{-2}$ (reach Mayrhofen) to $2125 \text{ ind} \cdot \text{m}^{-2}$ (reach Hart) and $11.14 \text{ g} \cdot \text{m}^{-2}$ to $3.58 \text{ g} \cdot \text{m}^{-2}$ respectively (Fig. 3).

Fig. 5 compares the habitat suitability for selected taxa in terms of depth-averaged flow velocity. Both examples of hydraulic suitability of two indicative taxa, the Ephemeroptera genus *Rhithrogena* and the Trichoptera species *Allogamus auricollis*, are based on the data series of the hydrologically unaffected investigation reach Mayrhofen and the affected reaches Ramsau and Hart, considering the theoretical maximum current velocities during hydropeaking to reflect the variability in habitat preferences of macroinvertebrates. The abundance of *Rhithrogena* increases significantly from averaged 53 (Mayrhofen) to $855 \text{ ind} \cdot \text{m}^{-2}$ (Hart), in contrast *Allogamus auricollis* decreases from 1504 to $155 \text{ ind} \cdot \text{m}^{-2}$ (Fig. 4).

Based on predictive habitat modelling, Fig. 6 presents the variability in habitat distribution due to hydropeaking of the selected taxa *Rhithrogena* sp. and *Allogamus auricollis*. The predictive habitat modelling shows two different patterns of physically based (flow velocity) response concerning the artificial fluctuating flows and different target species. For *Allogamus auricollis* we calculated a decrease in Weighted

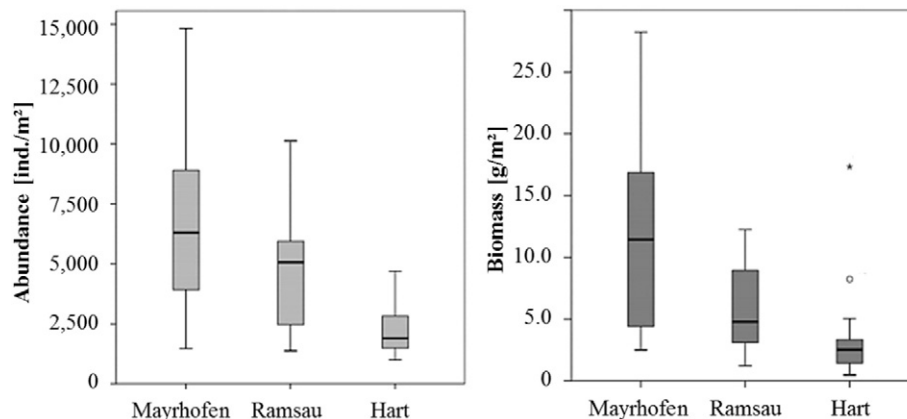


Fig. 3. Abundance (left) and biomass (right) per sampling site; modified Box samples ($n = 18$ per sampling site).

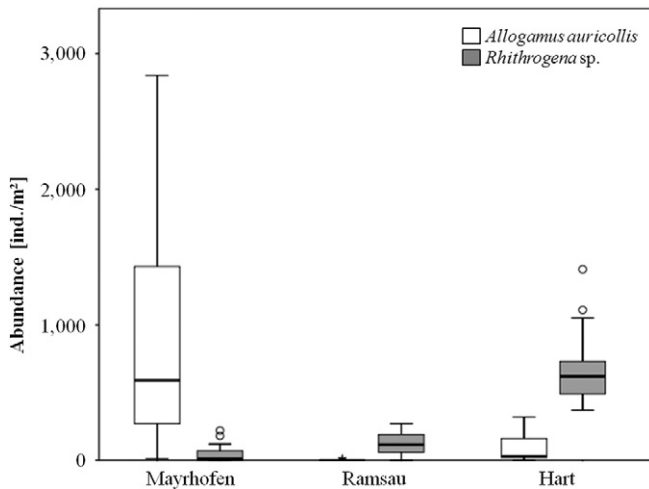


Fig. 4. Abundance of *Allogamus auricollis* (extreme values of 6730 ind·m⁻² and 8760 ind·m⁻² respectively at Mayrhofen are not displayed) and *Rhithrogena* sp. per sampling site; modified Box samples (n = 18 per sampling site).

Usable Areas (WUA) from 1519 m² to 540 m², resulting in a decrease from 22.1% to 5% in relation to the entire wetted area, especially for the suitable habitat at the gravel bar (Fig. 6c). For *Rhithrogena* sp., however, we documented an increase of WUA in relation to the determined depth-averaged flow velocity suitability (Fig. 5). In total we quantified an increase from 1604 m² (23.3%) to 6070 m² (56.2%).

5. Discussion

Assessment systems based on Biological Quality Elements aim to detect and measure anthropogenic influences on biota with standardized field methods and data analyses. Various stressor-types have specific impacts on the fauna and multiple interactions between various stressors may have additive effects, which are identified by adequate methods. Hydromorphological degradation as straightening of river courses and bank fixation in combination with pulse releases reduces slow-flowing marginal habitats considerably thus affecting lentic organisms the most.

One of the main findings of the presented impact study is that the national status assessment is inappropriate to reflect the impact of pulse release at the investigated Ziller River. Although dramatic changes in diversity and abundance were documented with increasing discharge ratio between the variable base-flow and peak-flow, the national method for assessing the ecological status class (Detailed MZB method:

Ofenböck et al., 2010) failed to reveal the significant differences between reference and affected stretches.

We demonstrated significant differences in hydraulic preferences with regard to the ecological niches of two indicator taxa based on habitat suitability curves. The caddisfly *Allogamus auricollis*, preferring low-flow habitats up to 0.5 ms⁻¹ velocity, occurred in masses at the unaffected reference stretch (up to 8670 ind·m⁻²). As sessile filter feeder, it lives close to the surface and is particularly affected by hydrological changes (Waringer, 1989a, 1989b, 1993; Graf et al., 1992; Graf, 2013). This species – representing a considerable energy source for predatory fish (Pletterbauer et al., 2016) – showed population losses of almost 100% in the hydropeaking stretches due to habitat loss during pulse release. Fig. 6a shows the modelled potential habitat without pulse release and Fig. 6b the actual situation, which – although modelled – fits very well with our field experience. In contrast, the mayfly genus *Rhithrogena* prefers zones of high velocity in the lotic river parts, providing a habitat preference optimum in current velocities around 1.5 ms⁻¹, showed a significant increase in individual numbers in affected stretches, illustrated by the modelled habitat for both scenarios (Fig. 6b, c). Differences in hydraulic preferences of both model taxa are in consensus with literature (e.g. Schmedtje, 1995; Caballita et al., 2013), showing that rheobiont taxa tolerate or even benefit in hydropeaking stretches compared to stagnophilic species.

We therefore suggest modifying sampling methods as well as assessment systems stressor-specific. In the case of hydropeaking, the focus should be laid on sensitive habitats as riverbanks and its lentic colonisers as *Allogamus auricollis*. Certain metrics as for example the abundance of stagnophilic organisms or the ratio of stagnophilic to rheobiont communities have the indicative power to identify impacts in terms of hydromorphology and should be implemented in the multi-metric index calculation for the module General Degradation in the existing assessment system of Austria. The analysed taxa are furthermore appropriate biological indicators for defining and identifying different ecological and specifically hydraulic stress-relevant niches in Alpine rivers.

6. Conclusions

Our results clearly show that 1) habitats of stagnophilic macroinvertebrate taxa are minimized in channelized stretches affected by hydropeaking, leading to heavy losses for populations, becoming apparent in significant decrease of total individual numbers and biomass for many taxa. 2) The minor respond of the ecological status class in affected stretches by applying the WFD compliant national assessment method for macroinvertebrates owes to the tolerance of rheobiont or rheophilic taxa commonly classified as indicators for good conditions

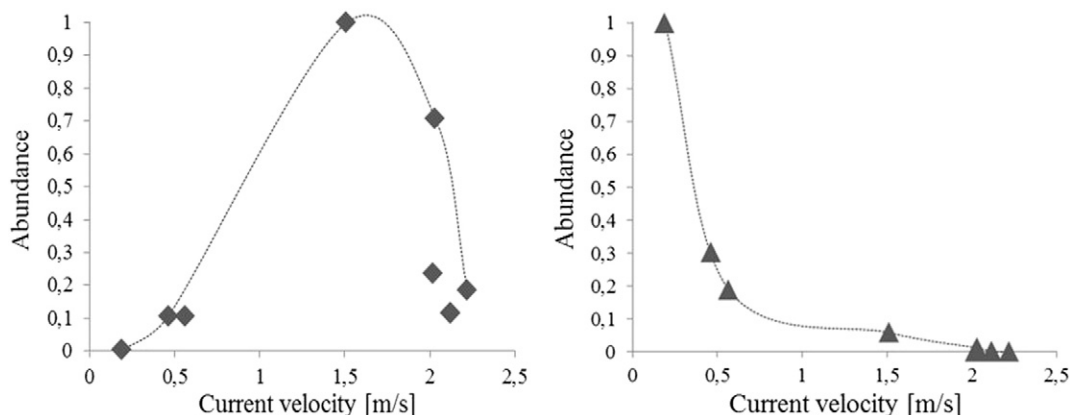


Fig. 5. Habitat suitability curves of *Rhithrogena* sp. (left) and *Allogamus auricollis* (right); box samples (current velocity is averaged per transect).

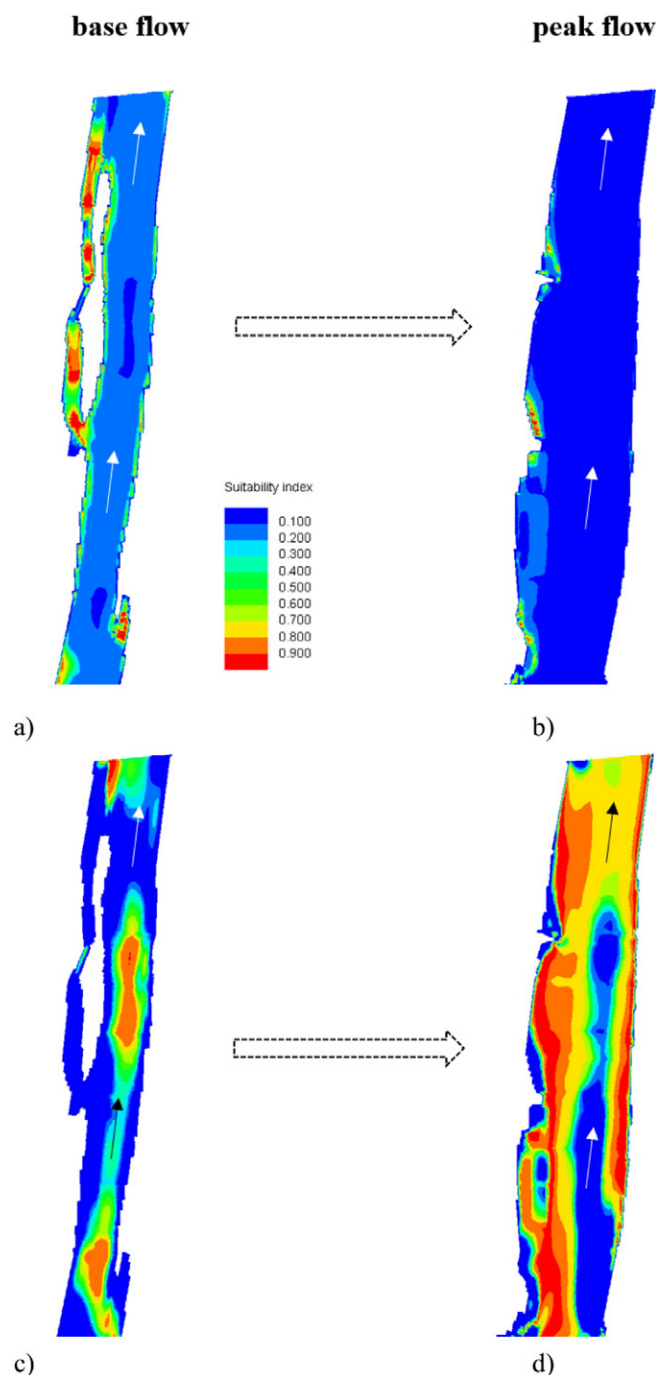


Fig. 6. Changes in habitat distribution for reach 1 (Ramsau) at the Ziller river based on unsteady two-dimensional predictive habitat modelling for *Allogamus auricollis* (a, b) and *Rhitrogena* sp. (c, d) due to hydropeaking; $Q_{\text{base}} = 3.95 \text{ m}^3 \text{ s}^{-1}$; $Q_{\text{peak}} = 74.72 \text{ m}^3 \text{ s}^{-1}$.

regarding saprobity or degradation score. Furthermore, abundance and biomass are not involved in the multi-metric calculation (Ofenböck et al., 2010) and do not reflect the effects of hydropeaking events although a significant decrease was obvious. 3) The development of a stressor-specific sampling design is required as the MHS largely ignores vulnerable habitats. 4) Habitat suitability of selected species turned out as efficient expertise for impact assessment and mitigation measure design in terms of predictive habitat modelling. The hydraulic stress analysis provides expertise on the resistance of certain taxa in terms of peak-flow (increasing risk of drift). Abundance values of these taxa are valuable indicators and are appropriate in a modular assessment system.

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